

Topic to be cover: Hypothesis testing of comparison between 2 means when population variance is known

Statistical Inference for Comparison Between Two Means (σ_1^2 and σ_2^2 Unknown):

In statistical inference, when population variances are **unknown**, we use the **t-test** to compare two population means. This method is especially useful in real-world situations where population parameters are rarely known.

For example, in finance, it can be used to compare the average returns of two portfolios, evaluate the performance of two mutual funds, or analyze differences in average interest rates between institutions.

By using sample data, we estimate and test whether the difference between the population means $\mu_1 - \mu_2$ is statistically significant.

Point Estimate:

The point estimate of the difference between two population means is:

$$\mu_1 - \mu_2 = \bar{X}_1 - \bar{X}_2$$

Interval Estimate (Confidence Interval):

The confidence interval depends on whether the population variances are assumed equal or unequal.

Case 1: Equal Variances (Pooled t-test)

When it is assumed that the two populations have equal variances $\sigma_1^2 = \sigma_2^2$, we use a **pooled variance**.

Pooled Variance:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Confidence Interval:

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\frac{\alpha}{2}, n_1 + n_2 - 2} \sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

Case 2: Unequal Variances (Welch's t-test)

When the population variances are not equal ($\sigma_1^2 \neq \sigma_2^2$), we do not pool them.

Confidence Interval:

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\frac{\alpha}{2}, df} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Where df is calculated as:

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}}$$

Book: Probability & statistics by Walpole page no 294

Question# 9.38 Two catalysts in a batch chemical process, are being compared for their effect on the output of the process reaction. A sample of 12 batches was prepared using catalyst 1, and a sample of 10 batches was prepared using catalyst 2. The 12 batches for which catalyst 1 was used in the reaction gave an average yield of 85 with a sample standard deviation of 4, and the 10 batches for which catalyst 2 was used gave an average yield of 81 and a sample standard deviation of 5. Find a 90% confidence interval for the difference between the population means, assuming that the populations are approximately normally distributed with equal variances.

$$s_1^2 = 4^2 = 16, s_2^2 = 5^2 = 25, n_1 = 12, n_2 = 10, \bar{X}_1 = 85, \bar{X}_2 = 81,$$

$$CL = 90, \frac{\alpha}{2} = 5\%, t_{\frac{\alpha}{2}, n_1+n_2-2} = t_{0.05, 12+10-2} = t_{0.05, 20} = 1.725$$

$$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2} = \frac{11 \cdot 16 + 9 \cdot 25}{12+10-2} = 20.05$$

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\frac{\alpha}{2}, n_1+n_2-2} \sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)} \Rightarrow (85 - 81) \pm 1.725 \sqrt{20.05 \left(\frac{1}{12} + \frac{1}{10}\right)}$$

$$\Rightarrow (0.6927 \text{ to } 7.3072)$$

We are 90% confident that difference in population mean lies between 0.6927 to 7.3072.

9.43 A taxi company is trying to decide whether to purchase brand *A* or brand *B* tires for its fleet of taxis. To estimate the difference in the two brands, an experiment is conducted using 12 of each brand. The tires are run until they wear out. The results are

Brand *A*: $\bar{x}_1 = 36,300$ kilometers,
 $s_1 = 5000$ kilometers.
Brand *B*: $\bar{x}_2 = 38,100$ kilometers,
 $s_2 = 6100$ kilometers.

Compute a 95% confidence interval for $\mu_A - \mu_B$ assuming the populations to be approximately normally distributed. You may not assume that the variances are equal.

Brand A: $\bar{x}_1 = 36,300$ kilometers,
 $s_1 = 5000$ kilometers.

Brand B: $\bar{x}_2 = 38,100$ kilometers,
 $s_2 = 6100$ kilometers.

$$s_1^2 = 5000^2, s_2^2 = 6100^2, n_1 = 12, n_2 = 12, \bar{X}_1 = 36300, \bar{X}_2 = 38100, CL = 95\%, \frac{\alpha}{2} = 2.5\% \\ = 0.025$$

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}} = \frac{\left(\frac{5000^2}{12} + \frac{6100^2}{12}\right)^2}{\frac{(5000^2)^2}{11} + \frac{(6100^2)^2}{11}} = 21.1839 \approx 22 t_{\frac{\alpha}{2}, df} = t_{0.025, 22} = 2.074$$

$$(36300 - 38100) \pm 2.074 \sqrt{\frac{5000^2}{12} + \frac{6100^2}{12}} \rightarrow (-6522.2418 \text{ to } 2922.2418)$$

We are 95% confident that difference in pop means will lie in between -6522.2418 to 2922.2418.

Hypothesis Testing of Difference Between Two Means

There are 6 basic steps:

1. State the claim & its opposite
2. Level of significance (α)
3. Test statistic
4. Critical region
5. Calculation
6. Conclusion

Test Statistic

Case 1: Equal Variances

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Case 2: Unequal Variances

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Decision Rule

The calculated t-value is compared with the critical value from the t-distribution based on degrees of freedom.

Roadmap of Comparison Between Two Means (t-test) for equal variance

Hypothesis	Test Statistics	Critical region
H₁: $\mu_1 - \mu_2 > a$ (when pop1 is greater, difference positive)	$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$ $s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$	$t > t_{\alpha, n_1 + n_2 - 2}$
H₁: $\mu_1 - \mu_2 < a$ (when pop 2 is greater, difference negative)		$t < -t_{\alpha, n_1 + n_2 - 2}$
H₁: $\mu_1 - \mu_2 \neq a$ (when both r different)		$t > t_{\alpha/2, n_1 + n_2 - 2}$ or $t < -t_{\alpha/2, n_1 + n_2 - 2}$

Roadmap of Comparison Between Two Means (t-test) for unequal variance

Hypothesis	Test Statistics	Critical region
H₁: $\mu_1 - \mu_2 > a$ (when pop1 is greater)	$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$ $df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{\left(\frac{s_1^2}{n_1} \right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2} \right)^2}{n_2 - 1}}$	$t > t_{\alpha, df}$
H₁: $\mu_1 - \mu_2 < a$ (when pop 2 is greater)		$t < -t_{\alpha, df}$
H₁: $\mu_1 - \mu_2 \neq a$ (when both r different)		$t > t_{\alpha/2, df}$ or $t < -t_{\alpha/2, df}$

Question#10.34 A study was made to determine if the subject matter in a physics course is better understood when a lab constitutes part of the course. Students were randomly selected to participate in either a 3-semester hour course without labs or a 4-semester-hour course with labs. In the section with labs, 11 students made an average grade of 85 with a standard deviation of 4.7, and in the section without labs, 17 students made an average grade of 79 with a standard deviation of 6.1. Would you say that the laboratory course increases the average grade by as much as 8 points? Use a P -value in your conclusion and assume the population to be approximately normally distributed with equal variances.

Excel, SPSS, panada, ➔ $P\text{-value} < 0.05$ result significant(reject H_0), $P\text{-value} > 0.05$ result are insignificant(accept H_0)

Step#1 $H_0: \mu_1 - \mu_2 \leq 8$

$H_1: \mu_1 - \mu_2 > 8$

Step#2 Level of significance: $\alpha = 0.05$

Step#3 Test Statistics

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Step#4 Critical region

$$t > t_{\alpha, n_1 + n_2 - 2}$$

$$t > t_{0.05, 11 + 17 - 2}$$

$$t > t_{0.05, 26}$$

$$t > 1.706$$

If critical region satisfies reject H_0

Step#5 Calculation:

In the section with labs, 11 students made an average grade of 85 with a standard deviation of 4.7, and in the section without labs, 17 students made an average grade of 79 with a standard deviation of 6.1. Would you say that the laboratory course increases the average grade by as much as 8 points?

$$s_1^2 = 4.7^2, s_2^2 = 6.1^2, n_1 = 11, n_2 = 17, \bar{X}_1 = 85, \bar{X}_2 = 79,$$

$$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2} = \frac{10 \cdot 4.7^2 + 16 \cdot 6.1^2}{11+17-2} = 31.3946$$

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{(85 - 79) - (8)}{\sqrt{31.3946 \left(\frac{1}{11} + \frac{1}{17} \right)}} = -0.9224$$

-0.9224 > 1.706 → doesn't satisfied

Step#6 Conclusion: As CR doesn't satisfied therefore there is no evidence reject H_0 , at 0.05 level of significance we conclude that lab course doesn't increase grade by 8 points.

Question# 10.39 The following data represent the running times of films produced by two motion-picture companies:

Company	Time (minutes)						
1	102	86	98	109	92		
2	81	165	97	134	92	87	114

Test the hypothesis that the average running time of films produced by company 2 exceeds the average running time of films produced by company 1 by 10 minutes against the one-sided alternative that the difference is less than 10 minutes. Use a 0.1 level of significance and assume the distributions of times to be approximately normal with unequal variances.

Step#1 $H_0: \mu_1 - \mu_2 \geq -10$

$H_1: \mu_1 - \mu_2 < -10$

Step#2 Level of significance: $\alpha = 0.1$

Step#3 Test Statistics

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{\left(\frac{s_1^2}{n_1} \right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2} \right)^2}{n_2 - 1}}$$

Step#4

Critical region

$t < t_{\alpha, df}$

Company	Time (minutes)						
1	102	86	98	109	92		
2	81	165	97	134	92	87	114

$$s_1^2 = 78.8, s_2^2 = 913.3333, n_1 = 5, n_2 = 7, \bar{X}_1 = 97.4, \bar{X}_2 = 110$$

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1-1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2-1}} = \frac{\left(\frac{78.8}{5} + \frac{913.3333}{7}\right)^2}{\frac{\left(\frac{78.8}{5}\right)^2}{4} + \frac{\left(\frac{913.3333}{7}\right)^2}{6}} = 7.3755 = 8$$

$T < -t_{0.10, 8}$

$T < -1.397$

If critical region satisfies reject H_0

Step#5 Calculation:

Company	Time (minutes)						
1	102	86	98	109	92		
2	81	165	97	134	92	87	114

$$s_1^2 = 78.8, s_2^2 = 913.3333, n_1 = 5, n_2 = 7, \bar{X}_1 = 97.4, \bar{X}_2 = 110$$

$$t = \frac{(97.4 - 110) - (-10)}{\sqrt{\frac{78.8}{5} + \frac{913.3333}{7}}} = -0.215$$

$-0.215 < -1.397 \rightarrow$ doesn't satisfied

Step#6 Conclusion: As CR doesn't satisfied therefore there is no evidence to reject H_0 at 0.10 level of significance we conclude that company 2 doesn't have larger waiting time than company 1

Question#3 To find out whether a new serum will arrest leukemia, 9 mice, all with an advanced stage of the disease, are selected. Five mice receive the treatment and 4 do not. Survival times, in years, from the time the experiment commenced are as follows:

Treatment	2.1	5.3	1.4	4.6	0.9
No Treatment	1.9	0.5	2.8	3.1	

At the 0.05 level of significance, can the serum be said to be effective? Assume the two populations to be normally distributed with equal variances.

Step#1 Ho:

H₁:

Step#2 Level of significance: $\alpha=0.01$

Step#3 Test Statistics

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$
$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Step#4

Critical region

$t > t_{\alpha, n_1 + n_2 - 2}$

$t < -t_{\alpha, n_1 + n_2 - 2}$

$t > t_{\alpha/2, n_1 + n_2 - 2}$ Or $t < -t_{\alpha/2, n_1 + n_2 - 2}$

If critical region satisfies reject Ho

Step#5 Calculation:

Treatment	2.1	5.3	1.4	4.6	0.9
No Treatment	1.9	0.5	2.8	3.1	

$s_1^2 =, s_2^2 =, n_1 =, n_2 =, \bar{X}_1 =, \bar{X}_2 =, CL =, \frac{\alpha}{2} =, t_{\frac{\alpha}{2}, n_1 + n_2 - 2}$

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$
$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Step#6 Conclusion:

Question# 4 Carbohydrates in Candies The number of grams of carbohydrates contained in 1-ounce servings of randomly selected chocolate and nonchocolate candy is listed here. Is there sufficient evidence to conclude that the difference in the means is significant? Use $\alpha=0.10$. assume variance are unequal.

Chocolate: 29, 25, 17, 36, 41, 25, 32, 29, 38, 34, 24, 27 & 29

Nonchocolate: 41, 41, 37, 29, 30, 38, 39, 10, 29, 55 & 29

Step#1 H_0 :

H_1 :

Step#2 Level of significance: $\alpha=0.01$

Step#3 Test Statistics

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}}$$

Step#4

Critical region

$t > t_{\alpha, df}$

$t < -t_{\alpha, df}$

$t > t_{\alpha/2, df}$ or $t < -t_{\alpha/2, df}$

If critical region satisfies reject H_0

Step#5 Calculation:

Chocolate: 29, 25, 17, 36, 41, 25, 32, 29, 38, 34, 24, 27 & 29

Nonchocolate: 41, 41, 37, 29, 30, 38, 39, 10, 29, 55 & 29

$$s_1^2 =, s_2^2 =, n_1 =, n_2 =, \bar{X}_1 =, \bar{X}_2 =, CL =, \frac{\alpha}{2} =, t_{\frac{\alpha}{2}, n_1 + n_2 - 2}$$

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}}$$

Step#6 Conclusion: